Northern University of Business and Technology, Khulna

Department of Computer Science and Engineering

CSE 4104 - Software Development III

Week 03 Submission: Software Requirement Specification (SRS)

Official Team Name	CSE4104-7F-T05
Section	7F
Project Title	AI Symptom Checker
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GitHub Repository	https://github.com/arnobpaul11/cse4104-7f-t05-ai-symptom-checker
Deployment Link	Not yet deployed (planned for Week 11: Vercel, Render, Supabase)
Submission Date	12 November 2026
Document Version	1.0 (based on the approved Week 02 Project Proposal)

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Submitted file	Content
CSE4104-7F-T05_SRS.pdf	This document
CSE4104-7F-T05_UseCaseDiagram.pdf	Use case diagram (actors, use cases, include/extend)
CSE4104-7F-T05_ERDiagram.pdf	ER diagram with primary and foreign keys (source: .dbml for dbdiagram.io)
CSE4104-7F-T05_ArchitectureDiagram.pdf	Initial system architecture diagram

1. Introduction

1.1 Project Title

AI Symptom Checker

1.2 Project Overview

The AI Symptom Checker is a secure web application in which a registered user describes symptoms in plain Bangla or English in a chat interface. An AI model asks adaptive follow-up questions, in the way a clinician takes a history, and the system returns a structured result: **possible conditions to consider**, a **suggested urgency level** (self-care, see a doctor soon, or emergency care), and **general next steps**. The user keeps a private health profile and symptom history, and can download a PDF summary to show a doctor. An admin dashboard provides usage statistics and management of users, disclaimers, and safety content.

The system is an **information and guidance tool only**. It does not give a medical diagnosis, prescribe medicine, or replace a licensed doctor.

1.3 Problem Statement

When people feel unwell, many first search the internet or ask friends and family, and the answers are often generic, contradictory, or alarming. In smaller cities and rural areas of Bangladesh, reaching a doctor quickly can be difficult, so people may delay care for a serious condition, take the wrong medicine, or visit a crowded hospital for a minor problem. Failing to recognise a warning sign such as chest pain, breathing difficulty, or sudden weakness can be dangerous, while unnecessary visits cost time and money. Existing symptom checkers are mostly built for international users, use fixed question flows, have no Bangla support, and keep no memory of earlier conversations. People need a quick, clear answer to one question: *how urgent is this, and what should I do next?*

1.4 Objectives

- Build a secure web application where registered users describe symptoms through a chat interface in Bangla or English.
- Use an AI model to ask adaptive follow-up questions instead of a fixed questionnaire.
- Produce a structured result: possible conditions, urgency level, and general next steps.
- Implement a two-layer red-flag system (rule-based check plus AI check) that immediately recommends emergency care for serious warning signs.
- Keep a private symptom history and health profile (age, gender, allergies, existing conditions) and use them as context for later assessments.
- Generate a downloadable PDF summary that a user can show to a doctor.
- Provide an admin dashboard for usage statistics and for managing users, disclaimer text, and safety content.
- Measure AI quality by testing at least 30 prepared symptom scenarios and recording the results.
- Deploy the system online and follow a professional workflow: GitHub branches, meaningful commits, documentation.

1.5 Scope of the Project

In scope	Out of scope
Web application (React.js frontend, Node.js/Express backend, PostgreSQL database) Authentication and two roles (User, Admin) AI chat intake with adaptive questions in Bangla and English Structured assessment with urgency level Rule-based and AI red-flag emergency alerts Health profile, private history, PDF summary Admin dashboard and content management Should-have: feedback, AI monitoring, personal analytics Could-have (only if time permits): body map, voice input, facility finder, audit logs	Medical diagnosis, prescriptions, or medicine dosage advice Replacing or connecting to a doctor, hospital, or ambulance service Real patient data (only test data is used during development) Native mobile apps (the web app is responsive instead) Payments, appointment booking, or telemedicine Regulatory approval as a medical device

1.6 Definitions and Acronyms

Term	Meaning
SRS	Software Requirement Specification
FR / NFR	Functional Requirement / Non-Functional Requirement
Urgency level	One of SELF_CARE, SEE_DOCTOR_SOON, EMERGENCY, assigned to every assessment
Red flag	A serious warning sign (for example chest pain, severe breathing difficulty, sudden weakness) that triggers an emergency recommendation
JWT	JSON Web Token, used for session authentication
RBAC	Role-based access control
Must / Should / Could	Feature priority levels from the Week 02 proposal (F1-F9 Must, F10-F11 Should, F12-F15 Could)

1.7 References

- CSE4104-7F-T05 Project Proposal (Week 02) and Initial Project Planning.
- CSE 4104 Student Handbook and Week 03 SRS instructions.
- Semigran, H. L., Linder, J. A., Gidengil, C., and Mehrotra, A. (2015). Evaluation of symptom checkers for self diagnosis and triage: audit study. BMJ, 351, h3480.

2. Functional Requirements

Each requirement is numbered and linked to a proposal feature (F1-F15) and a priority. Words such as *shall* describe mandatory behaviour for Must-have requirements.

Authentication and Roles (F1)

ID	Requirement	Feature	Priority
FR-01	Users shall register with full name, email, and password (minimum 8 characters), validated on client and server.	F1	Must
FR-02	The system shall reject registration with an email that already exists.	F1	Must
FR-03	Users shall log in with email and password; on success the system issues a JWT session token.	F1	Must
FR-04	Passwords shall be stored only as bcrypt hashes and shall never be returned by any API.	F1	Must
FR-05	The system shall enforce two roles, User and Admin; admin-only endpoints shall reject non-admin requests.	F1	Must
FR-06	Protected pages and APIs shall require a valid, unexpired token; users can log out, which ends the session on the client.	F1	Must

Health Profile (F5)

ID	Requirement	Feature	Priority
FR-07	Users shall view and update a health profile: date of birth, gender, allergies, existing conditions, current medications, and preferred language.	F5	Must
FR-08	The system shall include available profile data and a summary of past history as context in every AI prompt; empty fields are allowed.	F5	Must

AI Chat Symptom Intake and Language (F2, F8)

ID	Requirement	Feature	Priority
FR-09	Users shall start a new symptom chat session from the dashboard.	F2	Must
FR-10	Users shall describe symptoms in free text in Bangla or English.	F2, F8	Must
FR-11	Users shall choose the chat language, and the AI shall reply in the chosen language.	F8	Must
FR-12	The AI shall ask adaptive follow-up questions (for example duration, severity, associated symptoms) based on previous answers.	F2	Must
FR-13	The system shall limit a session to a maximum number of follow-up rounds (default 5), and the user may request the result earlier.	F2	Must
FR-14	All messages of a session shall be stored and linked to the session and the user.	F2	Must

Red-Flag Emergency Safety (F4)

ID	Requirement	Feature	Priority
FR-15	Every user message shall be checked by a rule-based red-flag check on the backend before it is sent to the AI.	F4	Must
FR-16	If a red flag is detected, the system shall immediately show an emergency alert recommending urgent medical care, without waiting for the AI.	F4	Must
FR-17	The AI response shall also be checked for red flags; a flag from either the rule layer or the AI sets the urgency level to EMERGENCY.	F4	Must
FR-18	Admins shall add, edit, enable, and disable red-flag rules (pattern, language, description).	F4, F9	Must

Structured Assessment (F3)

ID	Requirement	Feature	Priority
FR-19	The system shall produce a structured assessment containing possible conditions with short explanations, an urgency level (SELF_CARE, SEE_DOCTOR_SOON, EMERGENCY), and general next steps.	F3	Must
FR-20	The AI shall return the assessment as JSON; the backend shall validate it against a schema and, if invalid, retry once and then show a safe fallback message.	F3	Must
FR-21	Results shall be worded as 'possible conditions to consider' and shall not state a definitive diagnosis or give medicine dosages.	F3	Must
FR-22	A disclaimer shall be shown at the start of every chat and on every result.	F3	Must
FR-23	Each completed assessment shall be saved in the database and linked to its session.	F3	Must

Symptom History (F6)

ID	Requirement	Feature	Priority
FR-24	Users shall view a list of past sessions showing date, title, and urgency level.	F6	Must
FR-25	Users shall open a past session to read its conversation and result.	F6	Must
FR-26	Users shall delete a single session or all of their history; deletion is permanent.	F6	Must
FR-27	A user's history shall be visible only to that user.	F6	Must

PDF Health Summary (F7)

ID	Requirement	Feature	Priority
FR-28	Users shall download a PDF summary of a session containing profile basics, reported symptoms, answers, possible conditions, urgency level, next steps, date, and the disclaimer.	F7	Must
FR-29	The PDF generator shall render both English and Bangla text correctly.	F7, F8	Must

Admin Dashboard (F9)

ID	Requirement	Feature	Priority
FR-30	Admins shall view statistics: total users, total assessments, urgency distribution, red-flag count, and daily usage.	F9	Must
FR-31	Admins shall search, deactivate, and reactivate user accounts.	F9	Must
FR-32	Admins shall not be able to read individual users' symptom conversations; they see only account data and aggregate statistics.	F9	Must
FR-33	Admins shall edit the disclaimer and safety notice text in Bangla and English.	F9	Must

Should-have Features (F10, F11)

ID	Requirement	Feature	Priority
FR-34	Users shall rate an assessment with thumbs up or down and an optional comment.	F10	Should
FR-35	Admins shall view feedback and flagged AI failures to improve prompts.	F10	Should
FR-36	Users shall view charts of their symptom frequency and history patterns.	F11	Should

Could-have Features (F12-F15)

ID	Requirement	Feature	Priority
FR-37	Users may select the body region where a symptom occurs (interactive body map).	F12	Could
FR-38	Users may describe symptoms by voice using speech-to-text.	F13	Could
FR-39	Users may find nearby hospitals and clinics.	F14	Could
FR-40	The system may record admin actions in audit logs and send reminders about saved reports.	F15	Could

3. Non-Functional Requirements

Performance

ID	Requirement
NFR-01	Pages shall load within 3 seconds on a normal broadband connection.
NFR-02	Non-AI API calls shall respond within 1 second for at least 95% of requests.
NFR-03	An AI reply shall normally arrive within 10 seconds; the UI shall show a loading indicator and a timeout message after 30 seconds.

Security

ID	Requirement
NFR-04	User passwords shall be hashed with bcrypt (at least 10 salt rounds).
NFR-05	All traffic shall use HTTPS in the deployed system.
NFR-06	JWT tokens shall expire (default 24 hours) and be signed with a secret kept in an environment variable.
NFR-07	All input shall be validated and sanitised, and database queries shall be parameterised to prevent injection and XSS.
NFR-08	Rate limiting shall protect login and AI endpoints against abuse.
NFR-09	API keys and credentials shall exist only in environment variables, never in the frontend or the GitHub repository; AI APIs are called only from the backend.

Privacy and Data Protection

ID	Requirement
NFR-10	Each user's data shall be isolated; every query for health data shall be filtered by the authenticated user.
NFR-11	Name and email shall not be sent to the AI provider; only the symptom text, age, gender, and relevant conditions are sent.
NFR-12	Only test data shall be used during development; no real patient data.
NFR-13	Deleting history shall remove the related rows from the database.

Usability and Localisation

ID	Requirement
NFR-14	The interface shall be responsive on desktop, tablet, and mobile.
NFR-15	Bangla and English text shall display correctly (UTF-8 and a Bangla-capable font).
NFR-16	Every screen shall have loading, error, and validation messages written in plain language, with sufficient colour contrast and form labels.

Reliability and Availability

ID	Requirement
NFR-17	If the AI provider fails or times out, the system shall show a clear error and keep the session; the rule-based red-flag layer shall still work.
NFR-18	The deployed system shall target at least 95% availability during the evaluation period (free-tier cold starts are tolerated).

Safety and AI Quality

ID	Requirement
NFR-19	In testing, 100% of prepared red-flag scenarios shall produce an emergency recommendation.
NFR-20	At least 30 AI test scenarios shall be recorded with results in the testing report.
NFR-21	No result shall be presented without the disclaimer.

Maintainability, Scalability, Compatibility

ID	Requirement
NFR-22	The AI integration shall be a provider-independent module so Gemini or OpenAI can be swapped without changing other code.
NFR-23	The code shall be modular (frontend, backend, AI service), follow the GitHub branch workflow, and be documented in the README.
NFR-24	The backend shall be stateless so that it can scale; the demo shall support at least 50 concurrent users.
NFR-25	The application shall work on the latest versions of Chrome, Edge, Firefox, and mobile Safari/Chrome.

4. User Roles

Role	Description	Responsibilities and permissions
Guest	Visitor who is not logged in.	View landing page and disclaimer; register; log in.
User (general user)	Registered person who seeks guidance about symptoms.	Manage own profile; start symptom chats in Bangla or English; answer follow-up questions; view results and emergency alerts; view and delete own history; download PDF summaries; give feedback; view personal analytics. Cannot access admin features or other users' data.
Admin	Course project administrator who maintains the system.	Log in to admin dashboard; view usage statistics; manage (search, deactivate, reactivate) users; manage disclaimer and safety content; manage red-flag rules; review feedback and AI failures. Cannot read individual symptom conversations.
AI Service (external system)	Google Gemini API (primary) or OpenAI API (fallback).	Generate adaptive follow-up questions and the structured assessment from prompts sent by the backend. It has no direct access to the database or to users.

A Moderator role is not used because the system has no public or community content to review.

Capability	Guest	User	Admin
Register / Log in	Yes	-	-
Manage own profile	-	Yes	Yes (own)
Symptom chat and assessment	-	Yes	-
View / delete own history, PDF	-	Yes	-
Give feedback	-	Yes	-
Admin dashboard, user management	-	-	Yes
Edit disclaimer, safety content, red-flag rules	-	-	Yes

5. Use Cases

Each use case lists the actor, the user's action, and the system response, with preconditions and exceptions. The Use Case Diagram file shows how they relate.

UC-01 Register Account

Actor	Guest
Precondition	User is not logged in.
Action	Opens the register page and enters full name, email, and password.
System Response	Validates input, checks that the email is unique, hashes the password with bcrypt, creates a User account, and redirects to login.
Alternate / Exception	Email already exists or password too short: shows a validation message and creates nothing.
Related requirements	FR-01, FR-02, FR-04

UC-02 Log In

Actor	User or Admin
Precondition	Account exists and is active.
Action	Enters email and password.
System Response	Validates credentials, issues a JWT, and opens the dashboard (User) or admin dashboard (Admin).
Alternate / Exception	Wrong credentials or deactivated account: shows an error; repeated failures are rate-limited.
Related requirements	FR-03, FR-05, FR-06

UC-03 Manage Health Profile

Actor	User
Precondition	Logged in.
Action	Opens Profile, edits date of birth, gender, allergies, conditions, medications, language, and saves.
System Response	Validates and saves the profile; confirms success; the data is used as context in future assessments.
Alternate / Exception	Invalid date or server error: shows an error and keeps the old data.
Related requirements	FR-07, FR-08

UC-04 Start Symptom Chat

Actor	User
Precondition	Logged in; disclaimer shown.
Action	Starts a new chat, selects language (UC-15), and types symptoms in free text.
System Response	Creates a session and stores the message. **Includes** the red-flag check; if none is found, builds a prompt (message + profile + history) and sends it to the AI service.
Alternate / Exception	Red flag found: UC-06 emergency alert is shown immediately. AI unavailable: shows an error and keeps the session.
Related requirements	FR-09, FR-10, FR-11, FR-14, FR-15, FR-16

UC-05 Answer Follow-up Questions and Receive Assessment

Actor	User, AI Service
Precondition	An open chat session exists.
Action	Reads each AI follow-up question and replies; or requests the result early.
System Response	AI generates the next adaptive question each round. After the final round the AI returns structured JSON; the backend validates it, saves the assessment and its possible conditions, and shows a result card.
Alternate / Exception	Invalid JSON: one retry, then a safe fallback message. Maximum rounds reached: result is generated automatically.
Related requirements	FR-12, FR-13, FR-17, FR-19, FR-20, FR-21, FR-22, FR-23

UC-06 View Assessment Result / Emergency Alert

Actor	User
Precondition	An assessment exists.
Action	Opens the result card.
System Response	Displays possible conditions with explanations, urgency level, next steps, and the disclaimer. If urgency is EMERGENCY, shows a prominent alert recommending immediate medical care.
Alternate / Exception	None.
Related requirements	FR-16, FR-19, FR-21, FR-22

UC-07 View Symptom History

Actor	User
Precondition	Logged in.
Action	Opens History and selects a past session.
System Response	Lists the user's own sessions (date, title, urgency) and shows the selected conversation and result.
Alternate / Exception	No history: shows an empty-state message.
Related requirements	FR-24, FR-25, FR-27

UC-08 Delete History

Actor	User
Precondition	Logged in; history exists.
Action	Chooses delete for one session or all sessions and confirms.
System Response	Asks for confirmation, then permanently deletes the sessions, messages, and assessments, and updates the list.
Alternate / Exception	User cancels: nothing is deleted.
Related requirements	FR-26, NFR-13

UC-09 Download PDF Summary

Actor	User
Precondition	A completed assessment exists.
Action	Clicks Download PDF on a result.
System Response	Generates a PDF (profile basics, symptoms, answers, conditions, urgency, next steps, date, disclaimer) and downloads it.
Alternate / Exception	Generation error: shows a message and lets the user retry.
Related requirements	FR-28, FR-29

UC-10 Give Feedback (Should)

Actor	User
Precondition	An assessment exists.
Action	Clicks thumbs up or down and optionally writes a comment.
System Response	Saves the feedback linked to the assessment and user and confirms.
Alternate / Exception	None.
Related requirements	FR-34

UC-11 View Usage Statistics

Actor	Admin
Precondition	Logged in as Admin.
Action	Opens the admin dashboard.
System Response	Displays totals, urgency distribution, red-flag count, and daily usage charts (aggregate data only).
Alternate / Exception	None.
Related requirements	FR-30, FR-32

UC-12 Manage Users

Actor	Admin
Precondition	Logged in as Admin.
Action	Searches users and deactivates or reactivates an account.
System Response	Updates the account status; a deactivated user cannot log in. No symptom content is shown.
Alternate / Exception	User not found: shows an empty result.
Related requirements	FR-31, FR-32

UC-13 Manage Disclaimer, Safety Content, and Red-Flag Rules

Actor	Admin
Precondition	Logged in as Admin.
Action	Edits the disclaimer or safety notice (Bangla and English) or adds, edits, or disables a red-flag rule.
System Response	Validates and saves; the new text and rules are used immediately by the chat and the red-flag check.
Alternate / Exception	Invalid or empty pattern: shows a validation error.
Related requirements	FR-18, FR-33

UC-14 Review Feedback and AI Failures (Should)

Actor	Admin
Precondition	Logged in as Admin.
Action	Opens the feedback view and filters by rating.
System Response	Lists feedback and flagged failures so prompts can be improved.
Alternate / Exception	None.
Related requirements	FR-35

UC-15 Select Language

Actor	User
Precondition	In a chat or profile.
Action	Chooses Bangla or English.
System Response	Switches interface labels and instructs the AI to reply in the chosen language; the choice is saved in the profile.
Alternate / Exception	None.
Related requirements	FR-11, NFR-15

6. Constraints, Assumptions, and Dependencies

Constraints

- The project must be completed within one 14-week semester by a team of four students, following the weekly deliverables.
- Free tiers of Vercel, Render, and Supabase are used; the AI API has quota and cost limits.
- The system is not a medical device and must always show the disclaimer.
- The stack is React.js, Node.js + Express.js, PostgreSQL, and the Gemini or OpenAI API.
- Only test data may be used; no real patient data.

Assumptions and Dependencies

- Users have a modern browser and an internet connection.
- The external AI provider is available; the provider-independent module and fallback provider reduce this risk.
- Users enter information honestly; the system cannot verify it.
- Third-party services (Supabase, Render, Vercel) remain available during evaluation.

7. Traceability Matrix

Feature	Description	Requirements	Use cases	Diagrams
F1	Authentication and roles	FR-01 to FR-06	UC-01, UC-02	Use Case, ER (users), Architecture
F2	AI chat symptom intake	FR-09 to FR-14	UC-04, UC-05	Use Case, ER (chat_sessions, messages), Architecture
F3	Structured assessment	FR-19 to FR-23	UC-05, UC-06	ER (assessments, assessment_conditions)
F4	Red-flag emergency alert	FR-15 to FR-18	UC-04, UC-06, UC-13	Use Case, ER (red_flag_rules), Architecture
F5	Profile-aware assessment	FR-07, FR-08	UC-03	ER (health_profiles)
F6	Symptom history	FR-24 to FR-27	UC-07, UC-08	ER (chat_sessions)
F7	PDF health summary	FR-28, FR-29	UC-09	Architecture (PDF Service)
F8	Bangla-English support	FR-10, FR-11, FR-29	UC-15	ER (users.preferred_language)
F9	Admin dashboard	FR-30 to FR-33	UC-11, UC-12, UC-13	Use Case, ER (system_content)
F10	Feedback and AI monitoring	FR-34, FR-35	UC-10, UC-14	ER (feedback)
F11	Personal analytics	FR-36	(extends UC-07)	Architecture (Recharts)
F12-F15	Could-have features	FR-37 to FR-40	Not in first release	Not modelled yet

Appendix A: Expected AI Result Format (JSON)

The backend asks the AI for the following structure and validates it before saving (FR-19, FR-20).

```
{
  "urgency_level": "SEE_DOCTOR_SOON",        // SELF_CARE | SEE_DOCTOR_SOON | EMERGENCY
  "red_flag": false,
  "possible_conditions": [
    { "name": "Gastritis", "explanation": "Burning pain after meals can ..." }
  ],
  "next_steps": ["Avoid spicy food", "See a doctor if pain lasts over 3 days"],
  "summary": "Short plain-language summary in the chosen language",
  "disclaimer_shown": true
}
```

Appendix B: Consistency with the Proposal

This SRS uses the same title, features (F1-F15), priorities, technology stack, and AI workflow as the Week 02 proposal. The diagrams were prepared from the requirements above: the Use Case Diagram covers UC-01 to UC-15, the ER Diagram has nine tables supporting FR-01 to FR-36, and the Architecture Diagram shows the numbered AI workflow (request, red-flag check, prompt, AI call, validation, save, display).